OPEN AGENT HACKATHON 2026 · GENAI.WORKS · TRACK: REASONING ARCHITECTURE

# Trade Memory

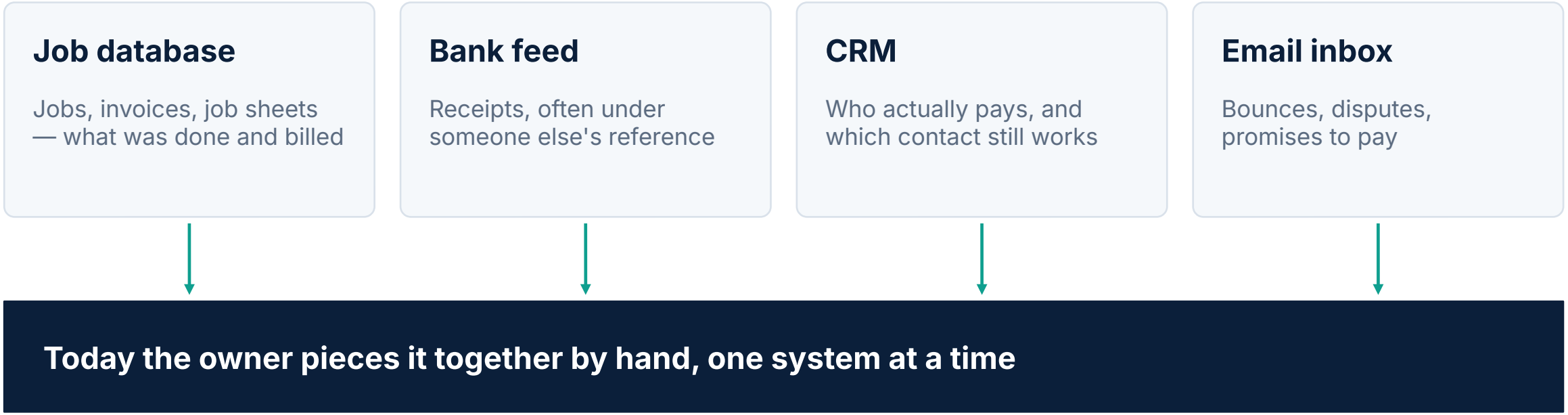
A credit-control agent for UK trades firms that shows its working — and never sends anything without your Approve.

**MN Labs Ltd** · Liverpool, UK · Founder: Michael Napier

github.com/MNLABSUK/trade-memory

THE PROBLEM

# Unpaid invoices hide across four systems that never talk




**A chaser that reads only one system sends the wrong message:** it chases invoices that were already paid, emails an inbox that bounces, and pushes on a dispute that the job sheet already settled. Each mistake costs cash flow or the customer relationship.

THE DEMO FIRM: HALDEN & ROOK (FICTIONAL)

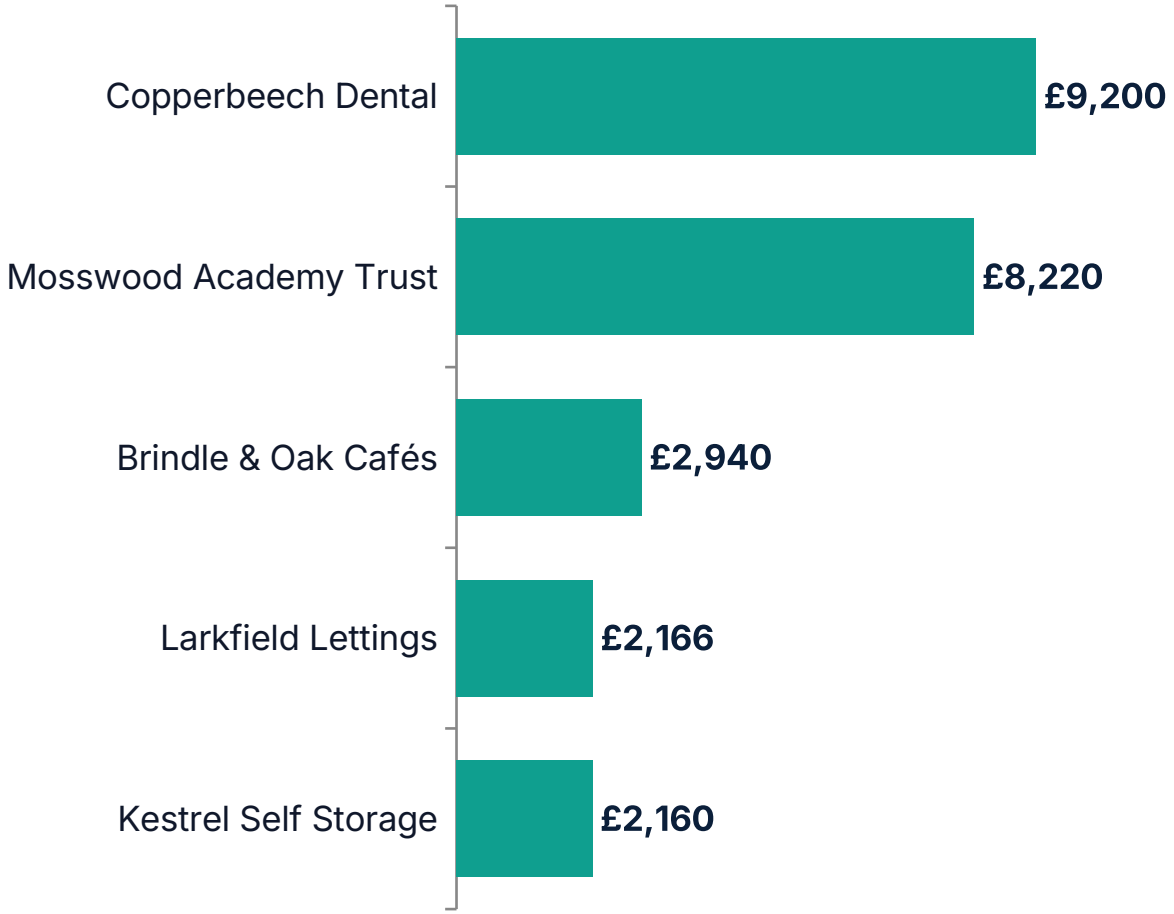
# £24,686 shows unpaid — each balance needs a different move

**£24,686**

unpaid across 12 invoices, computed from tm_ops.invoices

**Tom** runs Halden & Rook, a fictional electrical contractor. Every client, person and email in the demo data is made up.

**Five balances, five puzzles.** One is already paid. The other four need a new contact, a document, a PO or a phone call — not another email.



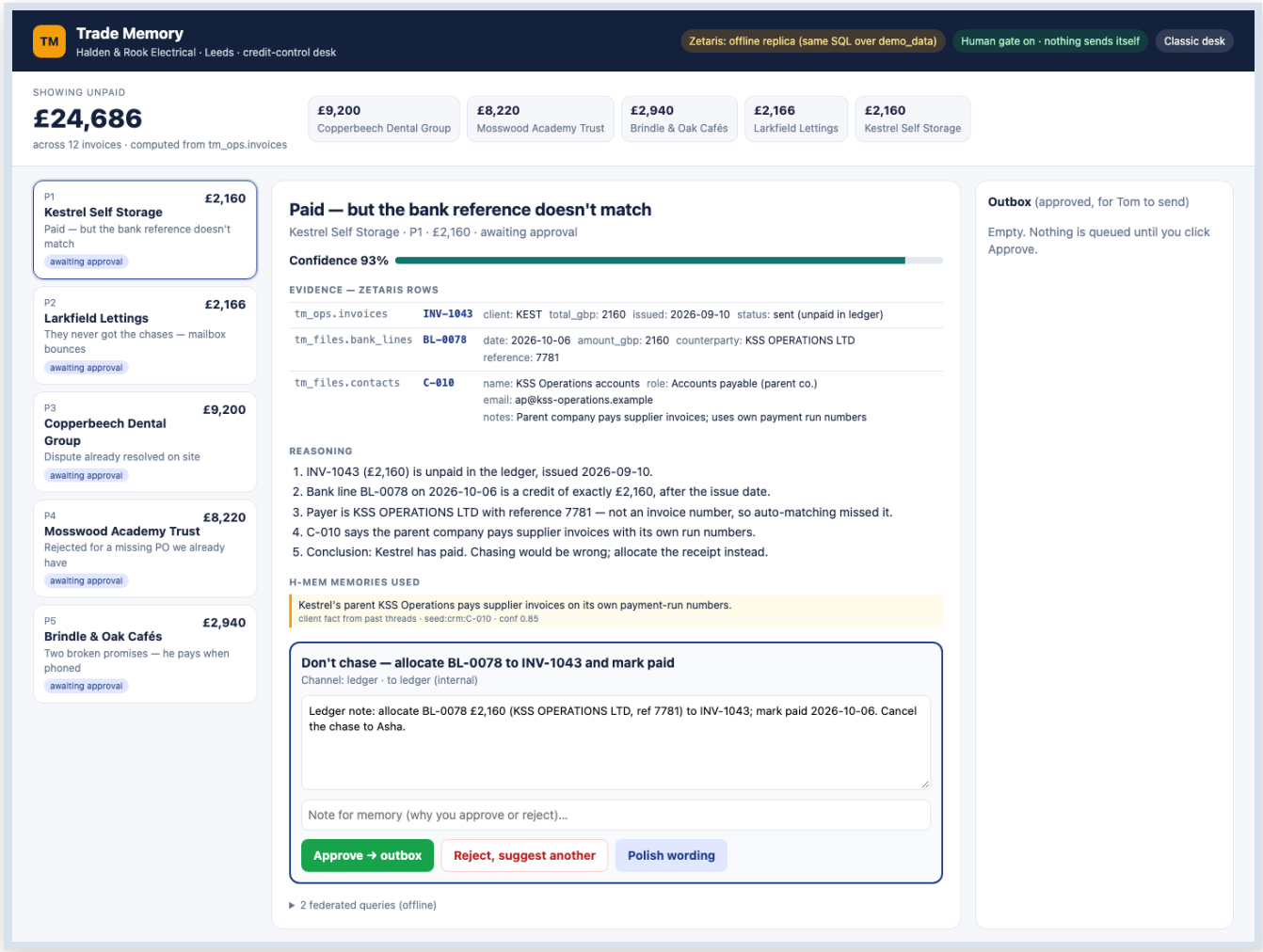
THE DEMO

# Five puzzles where the evidence beats a naive chaser

| Client | A naive chaser would… | What the federated evidence shows | Why card proposes |
|---|---|---|---|
| **Kestrel Self Storage** | Chase INV-1043 (£2,160) | Bank line BL-0078 paid by parent company KSS Operations, under its own reference (CRM C-010) | **Allocate the receipt. Don't chase** |
| **Larkfield Lettings** | Re-send to accounts@ | accounts@ bounced three times (MSG-0002/4/6); AP has moved to finance@ (C-002) | **Chase finance@ — Nadia Ferrow** |
| **Copperbeech Dental** | Chase £9,200 blind, or back off on the client's word | Client says a snag is open, but job sheet JS-216 shows the job closed on 2 Oct with sign-off | **Chase £9,200 on INV-1015, citing JS-216** |
| **Mosswood Academy Trust** | Chase INV-1018 (£8,220) | Rejected for a missing PO; job JOB-518 carries PO-MAT-77412 | **Reissue INV-1018 with PO-MAT-77412** |
| **Brindle & Oak Cafés** | Email again | Two missed promises (MSG-0011/12), no bank credit; he pays when phoned | **Phone Jonty today. If rejected → WhatsApp final notice** |

PRODUCT & UX

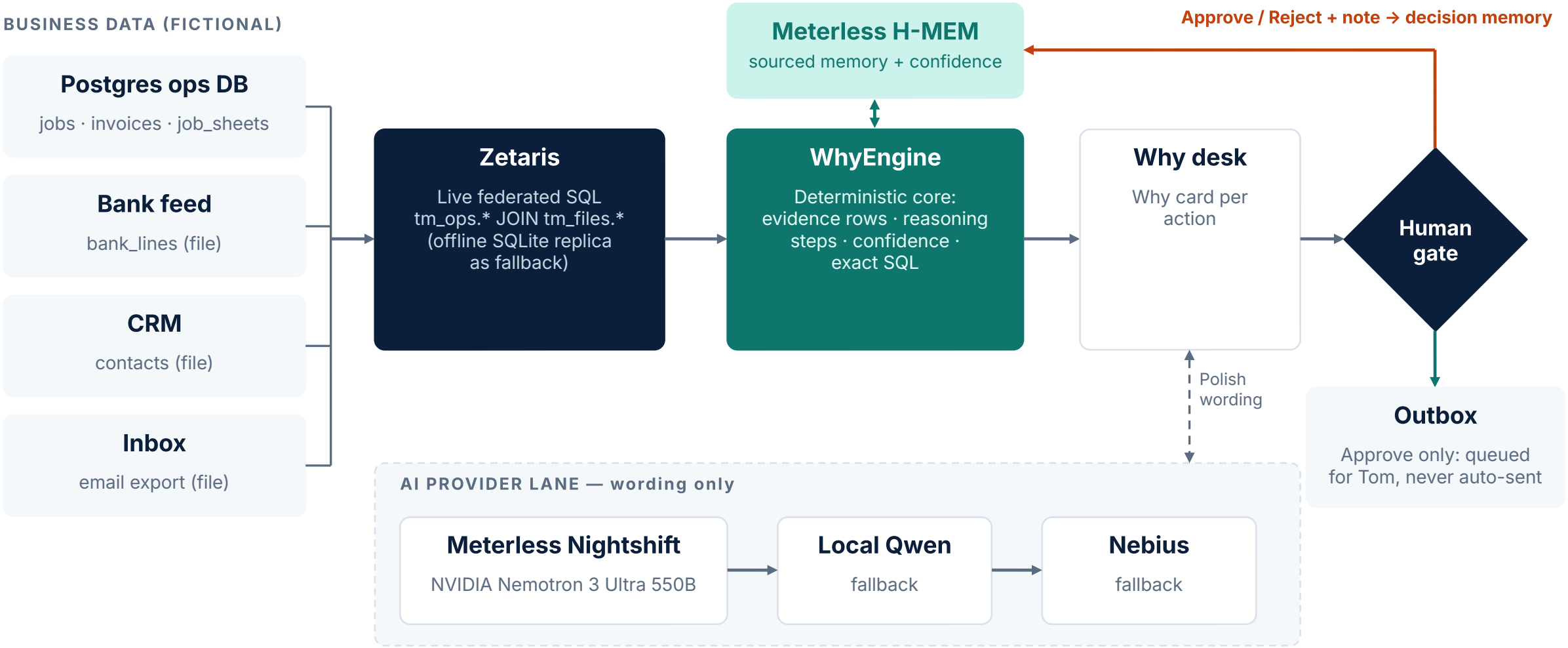
# Every action arrives as a Why card that cites its own evidence




1. **Source rows**
   Zetaris rows cited by ID: INV-1043, BL-0078, C-010
2. **Reasoning steps**
   Numbered, plain-English steps from the rows to the conclusion
3. **Confidence**
   93% here — a score, not a guess
4. **Memories used**
   H-MEM facts, each with its source and confidence
5. **Exact SQL**
   The federated queries behind the card, one click away
6. **Approve · Reject · Polish**
   Tom decides. Nothing leaves the desk on its own

TECHNICAL

# Architecture: federated facts in, gated decisions out




*Citation check: rewrite kept only if every invoice no., £ amount and PO survives*

REASONING ARCHITECTURE

# SQL decides the facts; the model only polishes wording

01

## Deterministic core

Card facts, the proposed action and the confidence score come from SQL rows and rules, never from an LLM. Every card is reproducible and testable.

**INV · PO · £ come from rows**

02

## A checked, swappable LLM

Polish wording rewrites a draft through Meterless Nightshift on NVIDIA Nemotron 3 Ultra 550B, falling back to local Qwen, then Nebius. A rewrite that drops or changes any invoice number, amount or PO is thrown away.

**Every INV / PO / £ is kept**

03

## Memory that learns

Meterless H-MEM stores each approval, rejection and client fact with its source and confidence. Approve moves the case on; Reject switches to the alternative action, lowers confidence and cites the rejection.

**Rejections are learned**

DEMO: BRINDLE & OAK CAFÉS (P5)

# Nothing sends without Approve — and rejections are learned

## Card proposes

Phone Jonty today — two promises missed, no bank credit, he pays when phoned

## Tom rejects, with a note

“Jonty is away this week, phone will not reach him”

## H-MEM stores the decision

A sourced decision memory for case P5

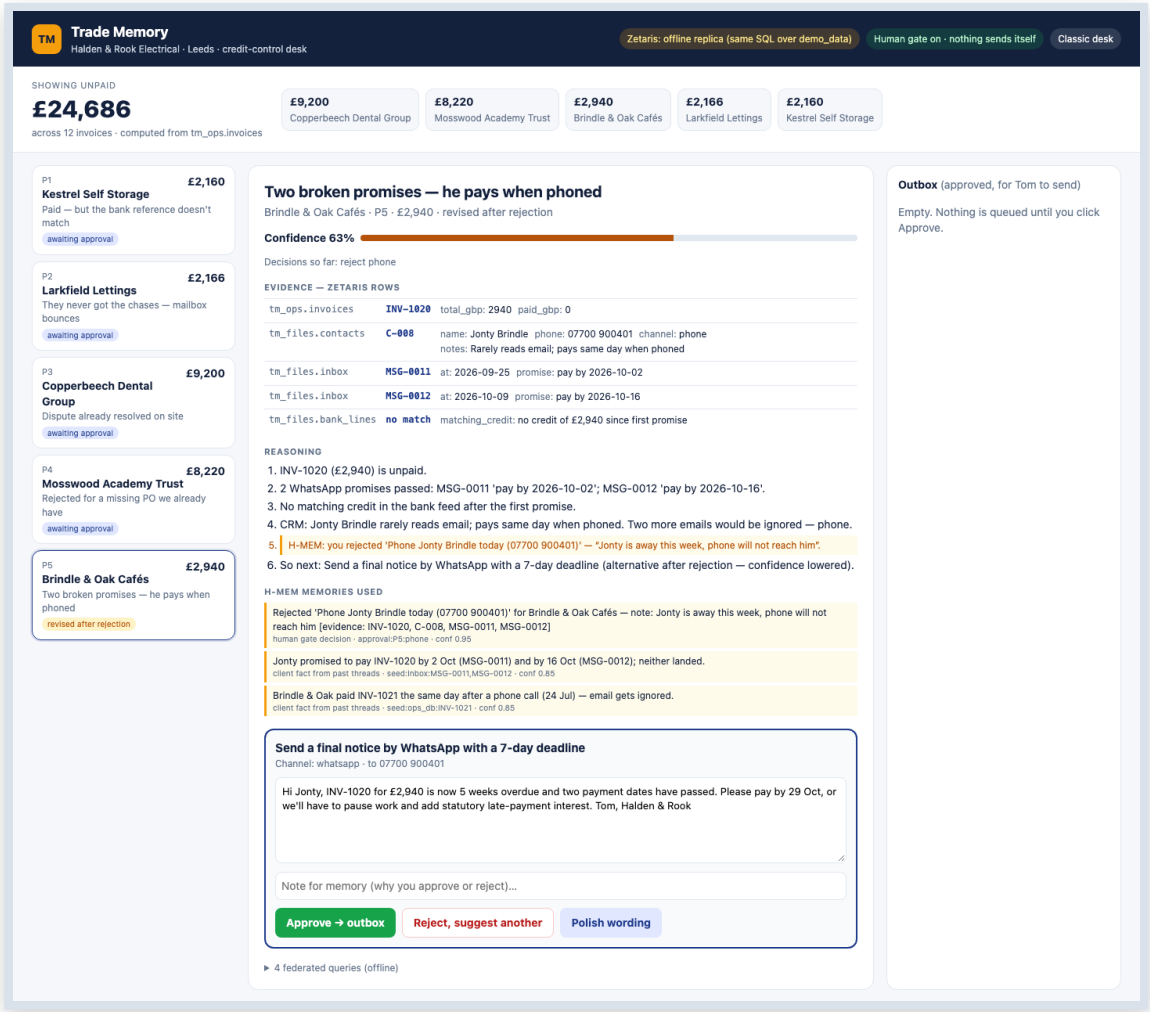
## Card re-plans and cites it

WhatsApp final notice with a 7-day deadline, still quoting INV-1020 and £2,940

CONFIDENCE

**0.86 → 0.63**

Lower confidence on the fallback action, shown openly on the card



SPONSOR TECHNOLOGY

# Sponsor tech does real work in every card — live, not mocked

## Zetaris

Live federated SQL across Postgres (tm_ops) and file sources (tm_files). Each card shows its queries, row counts and timings.

## Meterless Nightshift

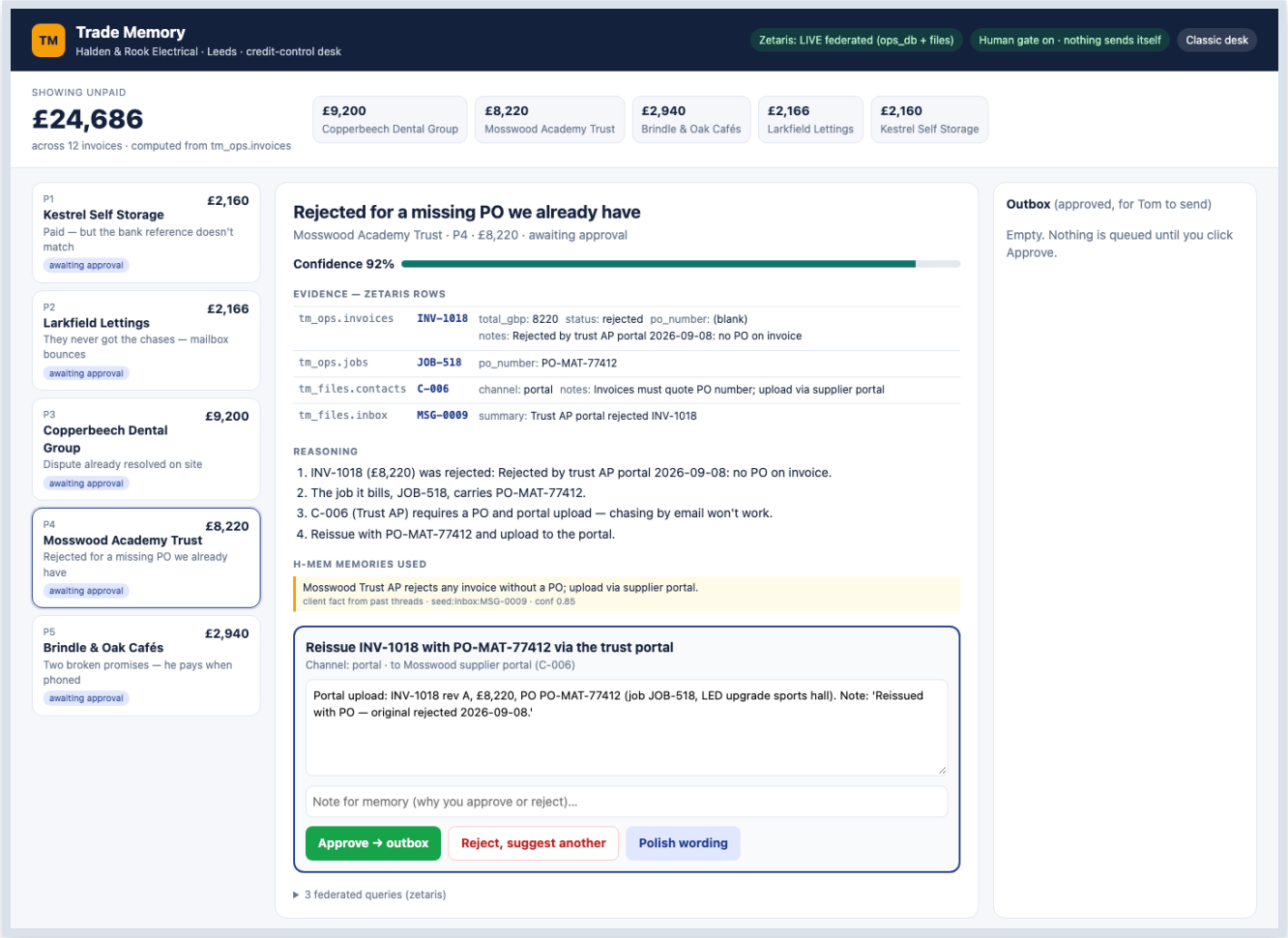
AI provider, tried first for every LLM call — serving NVIDIA Nemotron 3 Ultra 550B.

## Meterless H-MEM

Persistent, layered memory with provenance and confidence: decisions per case, client facts with source rows.

## NVIDIA Nemotron

Draft wording through Nightshift; local Qwen, then Nebius, as fallbacks.




*Screenshot: header reads "Zetaris: LIVE federated (ops_db + files)" on the Mosswood card*

TECHNICAL EXECUTION

# Built to be trusted: tested, honest about failure, and declared

**42**

## automated tests

Hermetic: dry-run LLM, temp H-MEM store and offline Zetaris, so they run anywhere with pytest -q.

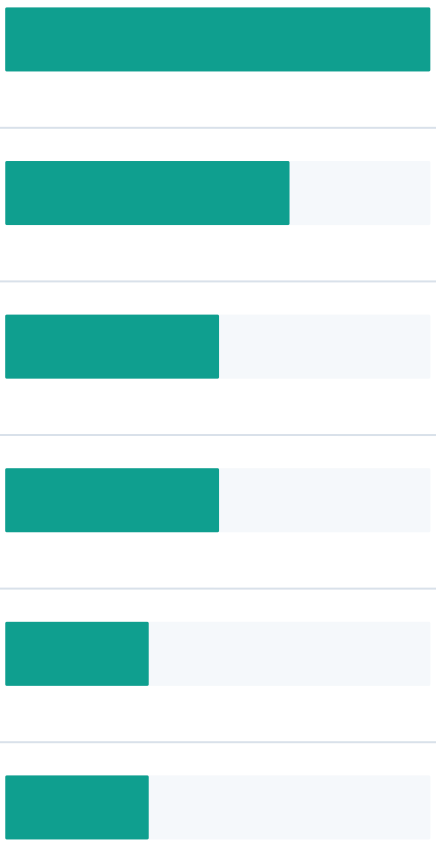
**Pre-existing components declared**

Baseline tag `pre-window-v0.3.0`, sanitised (fictional names, anonymised emails). All new work is reviewable in the diff from that tag.

| Failure mode | What Trade Memory does |
|---|---|
| Zetaris unreachable | Offline SQLite replica runs the same SQL; the header says which mode is live |
| LLM invents or drops a fact | Citation check keeps the rules draft. Facts never come from the LLM |
| Stale decision from a second tab | action_id check returns 409 Conflict |
| Wrong proposal | Reject + note goes into H-MEM; the alternative shows with lower confidence |
| Auto-send risk | No send path exists. Approve only writes to the outbox |

JUDGING CRITERIA

# How Trade Memory answers each judging criterion



| | | |
|---|---|---|
| **Impact** | 30 | A trades owner sees which of £24,686 to chase, who to contact and how — and avoids chasing paid invoices or dead inboxes |
| **Technical** | 20 | Live Zetaris federation, deterministic WhyEngine, H-MEM memory, provider fallbacks, 42 hermetic tests |
| **Innovation** | 15 | Evidence-cited Why cards with exact SQL; rejections become memories that change the next proposal |
| **Demo** | 15 | Five realistic puzzles end to end, including a live reject → re-plan (0.86 → 0.63) |
| **Product & UX** | 10 | One desk: Approve, Reject with a note, or Polish wording. Approved items queue for Tom |
| **Sponsor tech** | 10 | Zetaris, Meterless Nightshift, Meterless H-MEM and NVIDIA Nemotron each carry part of every card |

# Credit control that shows its working.

Federated evidence in. A cited, confident, human-approved action out. Every decision remembered.

**Code, demo video and screenshots**
github.com/MNLABSUK/trade-memory

**Run it**

```
pytest -q
uvicorn trade_memory.app:app --port 43151
```

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